

Loading the battery dataset

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%% STEP 1: Download and Load Battery Dataset
dataFile
= matlab.internal.examples.downloadSupportFile("predmaint","batteryagingdata/
singlecell/v1/singleCellLifeTimeData.zip");
unzip(dataFile);
load("singleCellLifeTimeData.mat");

%% STEP 2: Parse and Segment Data
bParser = batteryTestDataParser(data);
bParser.CurrentVariable = 'Current';
bParser.VoltageVariable = 'Voltage';
bParser.TimeVariable = 'DateTime';
bParser.CycleIndexVariable = 'Cycle_Index';
bParser.StepIndexVariable = 'Step_Index';
segmentedRawDataTable = segmentData(bParser);

%% STEP 3: Extract All Valid "Charge" Data for Cycle 1
chargingData = segmentedRawDataTable(...
    segmentedRawDataTable.Cycle_Index == 1 & ...
    segmentedRawDataTable.CyclingPhases == "Charge", :);
chargingData.TimeInSeconds = seconds(chargingData.DateTime -
chargingData.DateTime(1));
```

Compute the rate of voltage change

This quantifies the charging profile numerically rather than just eyeballing the plot.

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% Assumes t (time in seconds) and V (voltage) already extracted, e.g.:
t = chargingData.TimeInSeconds;
V = chargingData.Voltage;

vMax = 3.6;

% --- Identify time interval of 100% charge ---
i100 = find(V >= vMax, 1);
t100 = t(i100);
v100 = V(i100);

% --- Restrict analysis window to start-of-charge through 100% point ---
tPlot = t(1:i100);
vPlot = V(1:i100);

% --- Compute the rate of voltage change (dV/dt) across the whole window ---
% gradient() gives a numerical derivative at every sample point using
% central differences (forward/backward at the endpoints).
dVdt = gradient(vPlot, tPlot);

% --- Identify time interval of 80% charge ---
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v80Target = 0.80 * vMax;
i80 = find(vPlot >= v80Target, 1);
t80 = tPlot(i80);
v80 = vPlot(i80);           % actual voltage at/just past the 80% mark

% --- Identify time interval of 50% charge ---
v50Target = 0.50 * vMax;
i50 = find(vPlot >= v50Target, 1);
t50 = tPlot(i50);
v50 = vPlot(i50);

% --- Identify direction-change points (e.g., CC -> CV transition, noise) ---
% diff(sign(dVdt)) ~= 0 flags where the slope of V(t) flips sign
iDC = find(diff(sign(dVdt)) ~= 0) + 1;
tDC = tPlot(iDC);
vDC = vPlot(iDC);

% --- Compute dV/dt AT each key point of interest ---
dVdt_50 = dVdt(i50);
dVdt_80 = dVdt(i80);
dVdt_100 = dVdt(i100);

% --- Display a summary table of rate of voltage change at each interval ---
keyPointNames = {'50% Charge'; '80% Charge'; '100% Charge'};
keyTimes       = [t50; t80; t100];
keyVoltages    = [v50; v80; v100];
keyRates       = [dVdt_50; dVdt_80; dVdt_100];

summaryTable = table(keyPointNames, keyTimes, keyVoltages, keyRates, ...
    'VariableNames', {'ChargeLevel', 'Time_s', 'Voltage_V', 'dVdt_V_per_s'});

disp('--- Rate of Voltage Change at Key Charging Intervals ---')

```

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```
disp(summaryTable)
```

ChargeLevel	Time_s	Voltage_V	dVdt_V_per_s
{'50% Charge' }	0	2.0326	0.70662
{'80% Charge' }	2.6823	2.8826	0.048187
{'100% Charge' }	1246.5	3.6004	0.0014743

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fprintf('\nNote: dV/dt is highest early in charging (steep CC ramp) and\n')
```

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```
fprintf('decreases toward 0 as the battery approaches full charge (CV taper).\n')
```

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% Optional: Plot notable transitions, annotated with their dV/dt values
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figure('Name', 'Task 3.1');
plot(tPlot, vPlot, 'Color', [0.5, 0, 0.5]);
hold on;
```

```
plot(t50, v50, 'cs', 'MarkerFaceColor', 'c');
plot(t80, v80, 'ro', 'MarkerFaceColor', 'r');
plot(t100, v100, 'go', 'MarkerFaceColor', 'g');
plot(tDC, vDC, 'bo');
```

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% Annotate each key point with its instantaneous dV/dt value
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```
text(t50, v50, sprintf(' %.4f V/s', dVdt_50));
text(t80, v80, sprintf(' %.4f V/s', dVdt_80));
text(t100, v100, sprintf(' %.4f V/s', dVdt_100));
```

```
title('Voltage vs. Time (with Notable Transitions and dV/dt)');
xlabel('Time (s)');
ylabel('Voltage (V)');
legend('V(t)', '50% Charge', '80% Charge', '100% Charge', ...
'Direction Changes', 'Location', 'best');
grid on;
box on;
hold off;
```

